SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1103

COURSE OUTCOME COVERED
CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. (BL4)

Assessment Tool Weightage: 40%
Questions: 4
Total Marks: 40

Name / Reg No

GIRIJA B, 192311044

Code of Conduct

I, Girija B/ 192311044, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Girija B

Question 1: Book Bank Process

Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.

1. Identified Actors

- Student
- Administrator
- Central Computer
- Office Staff

2. Identified Use Cases

- Submit Student Details
- Verify Student Details
- Approve Student Request
- Request Books
- Issue Books

3. Use Case Diagram

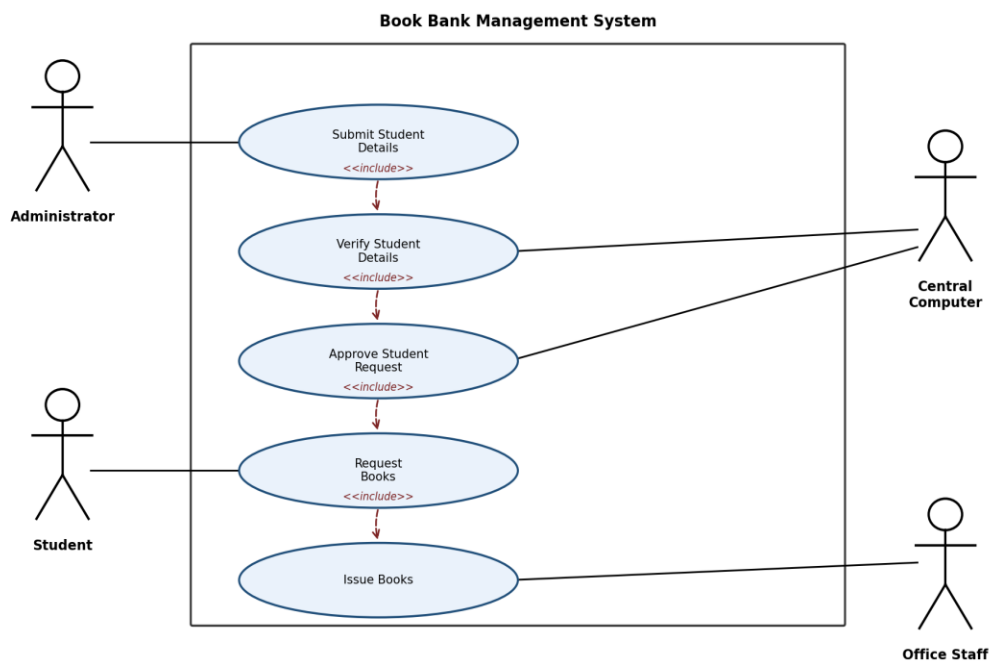

Fig 1.1: Use Case Diagram - Book Bank Management System

4. Process Flow / Explanation

- 1. The Administrator collects the student's identification and membership details at the book bank counter.

- 2. The Administrator submits these details to the Central Computer for verification.
- 3. The Central Computer checks the student's eligibility and record validity.
- 4. On successful verification, the Central Computer sends an approval to the Office.
- 5. The Student requests the required books at the Office.
- 6. The Office, upon seeing the approval, issues the requested books to the Student.

5. Key Relationships Used

- <<include>> — 'Submit Student Details' is required before 'Verify Student Details' can occur.
- <<include>> — 'Verify Student Details' must complete before 'Approve Student Request'.
- <<include>> — 'Approve Student Request' precedes and is required for 'Issue Books'.
- Association — Administrator interacts with Submit Details; Central Computer interacts with Verify/Approve; Office Staff interacts with Issue Books; Student interacts with Request Books.

Question 2: Exam Registration Portal

Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.

1. Identified Actors

- Candidate
- Administrator
- Central Computer
- Office Staff

2. Identified Use Cases

- Submit Candidate Details
- Verify Candidate Details
- Approve Candidate
- Register for Exam
- Issue Hall Ticket

3. Use Case Diagram

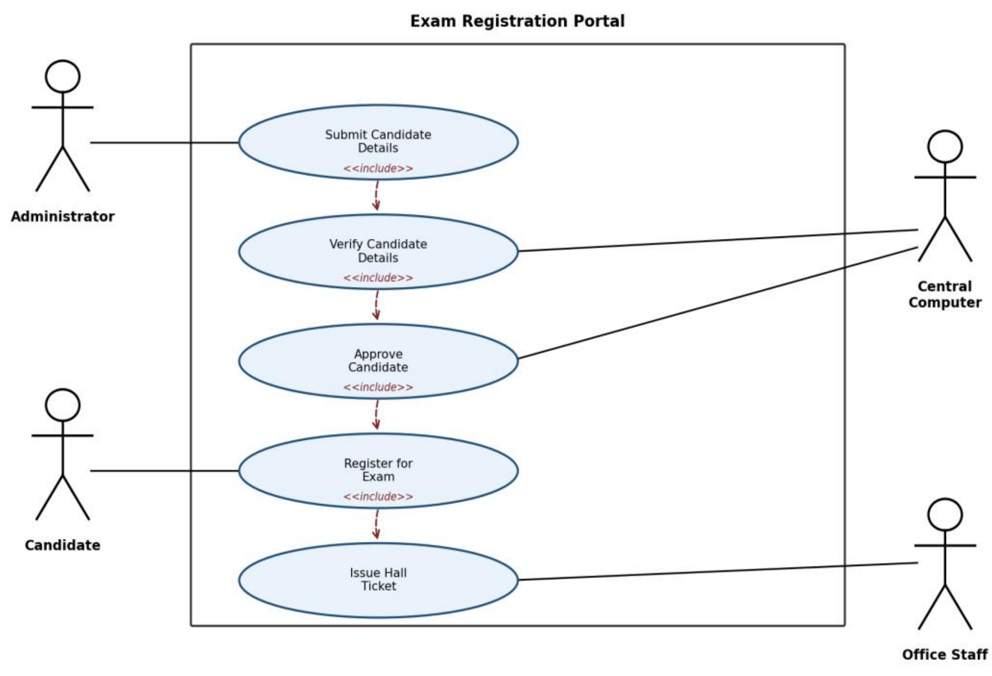

Fig 2.1: Use Case Diagram - Exam Registration Portal

4. Process Flow / Explanation

- 1. The Administrator collects and enters the candidate's registration details into the portal.

- 2. These details are forwarded to the Central Computer for verification.
- 3. The Central Computer validates the candidate's eligibility, fee payment status, and records.
- 4. Once verified, the Central Computer grants approval for the candidate's registration.
- 5. The Candidate registers for the desired examination through the portal.
- 6. The Office generates and issues the Hall Ticket to the approved Candidate.

5. Key Relationships Used

- <<include>> — 'Submit Candidate Details' must occur before 'Verify Candidate Details'.
- <<include>> — 'Verify Candidate Details' is required for 'Approve Candidate'.
- <<include>> — 'Register for Exam' precedes 'Issue Hall Ticket'.
- Association — Administrator submits details; Central Computer verifies/approves; Office Staff issues hall ticket; Candidate registers for exam.

Question 3: Stock Maintenance System

Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.

1. Identified Actors

- Customer
- Stock Dealer
- Central Stock System

2. Identified Use Cases

- Place Order
- Verify Order
- Deliver Items to Customer
- Update Stock Details

3. Use Case Diagram

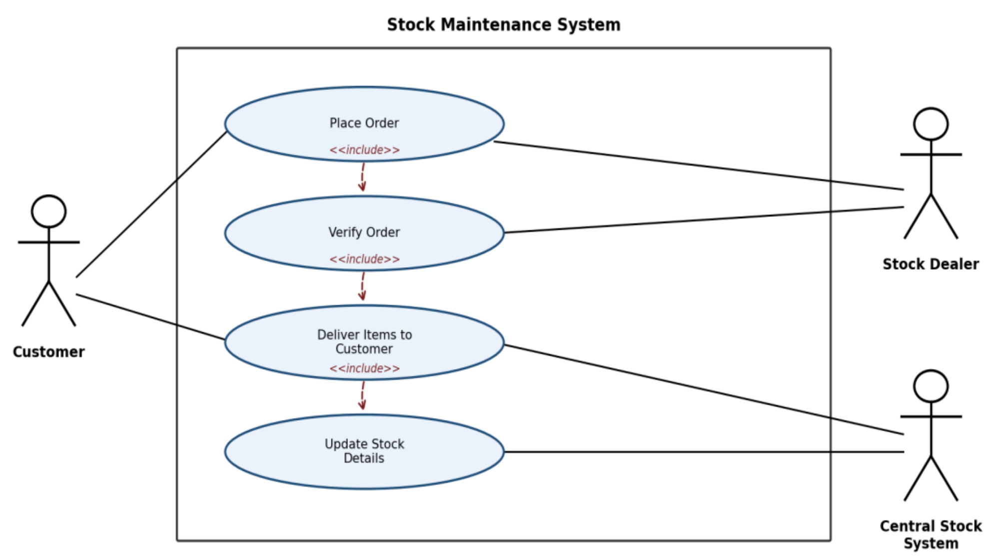

Fig 3.1: Use Case Diagram - Stock Maintenance System

4. Process Flow / Explanation

- 1. The Customer places an order for the required items through the Stock Dealer.
- 2. The Stock Dealer forwards the order to the Central Stock System for verification.
- 3. The Central Stock System checks item availability and confirms the order.
- 4. Once verified, the ordered items are delivered to the Customer.

- 5. The Central Stock System updates the stock details to reflect the items sold.

5. Key Relationships Used

- <<include>> — 'Place Order' must be completed before 'Verify Order'.
- <<include>> — 'Verify Order' is required before items can 'Deliver Items to Customer'.
- <<include>> — 'Deliver Items to Customer' triggers 'Update Stock Details'.
- Association — Customer places/receives order; Stock Dealer places and verifies order; Central Stock System verifies order and updates stock.

Question 4: Online Course Reservation System

Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

1. Identified Actors

- Student
- Central Management System
- University

2. Identified Use Cases

- Browse Available Courses
- Select Course
- Check Seat Availability
- Enroll Student in Course

3. Use Case Diagram

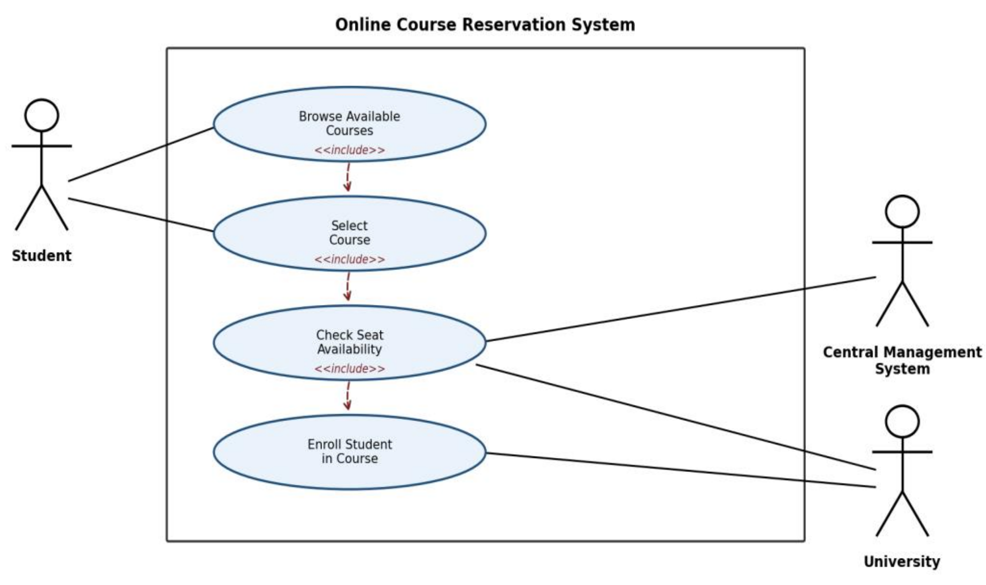

Fig 4.1: Use Case Diagram - Online Course Reservation System

4. Process Flow / Explanation

- 1. The Student browses the list of available courses through the Central Management System.
- 2. The Student selects the desired course.
- 3. The University checks whether a seat is available for the selected course.
- 4. If a seat is available, the Student is enrolled in the course.

- 5. If no seat is available, the enrollment request is rejected and the Student is notified.

5. Key Relationships Used

- <<include>> — 'Browse Available Courses' precedes 'Select Course'.
- <<include>> — 'Select Course' is required before 'Check Seat Availability'.
- <<include>> — 'Check Seat Availability' precedes 'Enroll Student in Course'.
- Association — Student browses/selects course; Central Management System checks seat availability; University enrolls the student.